SD model - Claire

```
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   4.0.2     v tibble    3.2.1
v lubridate 1.9.3     v tidyr     1.3.1
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
train <- read_csv("data-train.csv") |>
  mutate(
    C_moment_2 = R_moment_2 - R_moment_1^2,
    sd_R = sqrt(C_moment_2),
    Re = factor(Re),
    Fr = factor(Fr)
  )
```

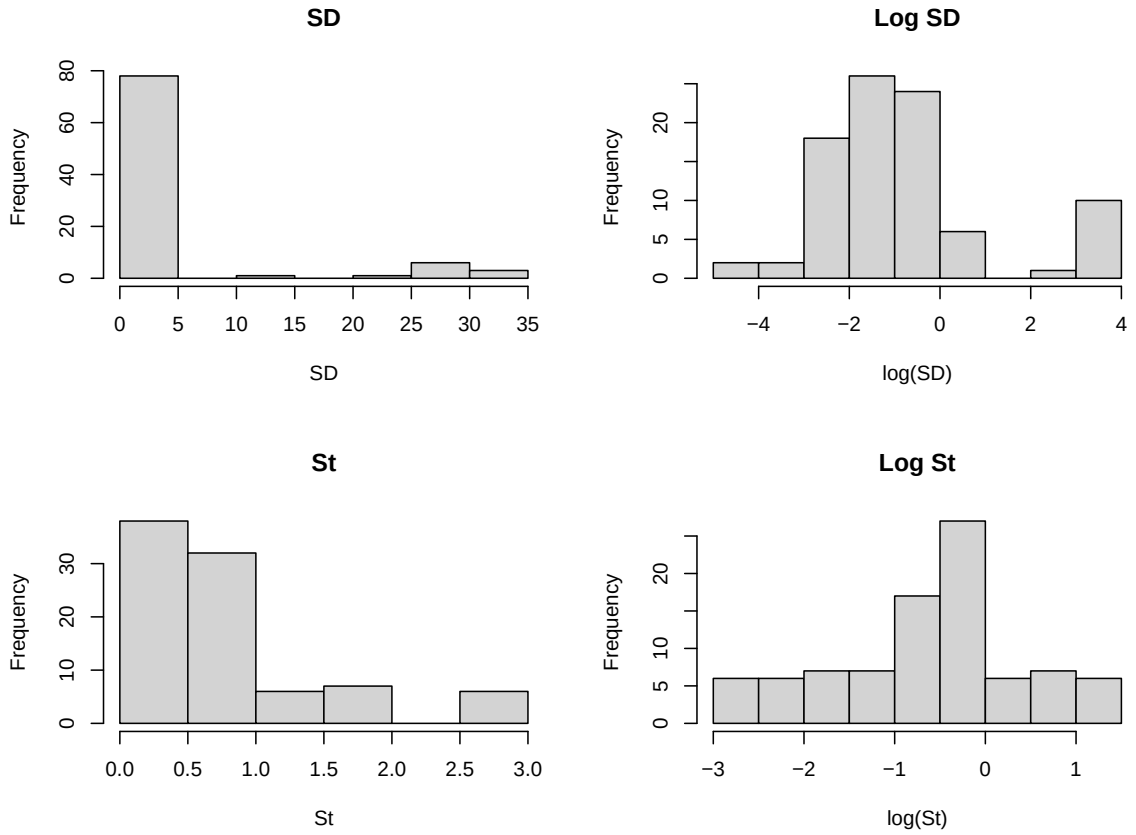
```
Rows: 89 Columns: 7
-- Column specification --------------------------------------------------------
Delimiter: ","
dbl (7): St, Re, Fr, R_moment_1, R_moment_2, R_moment_3, R_moment_4

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
stopifnot(all(is.finite(train$sd_R)), all(train$sd_R > 0),
          all(train$St > 0))
```

Explore SD and St

```r
par(mfrow = c(2, 2))
hist(train$sd_R, main = "SD", xlab = "SD")
hist(log(train$sd_R), main = "Log SD", xlab = "log(SD)")
hist(train$St, main = "St", xlab = "St")
hist(log(train$St), main = "Log St", xlab = "log(St)")
```




```r
par(mfrow = c(1, 1))
```

Compare log transformations using LOOCV

```
log_formulas <- list(
  no_logs = sd_R ~ St + Re + Fr,
  log_SD = log(sd_R) ~ St + Re + Fr,
  log_St = sd_R ~ log(St) + Re + Fr,
  log_both = log(sd_R) ~ log(St) + Re + Fr
)

evaluate_sd <- function(formula, data) {
  log_response <- grepl("log", deparse(formula[[2]]), fixed = TRUE)
  predictions <- numeric(nrow(data))
  for (i in seq_len(nrow(data))) {
    fit <- lm(formula, data = data[-i, ])
    pred <- predict(fit, newdata = data[i, , drop = FALSE],
                    rankdeficient = "NA")
    if (log_response) {
      correction <- mean(exp(residuals(fit)))
      pred <- exp(pred) * correction
    }
    predictions[i] <- pred
  }
  stopifnot(all(is.finite(predictions)))
  list(
    fit = lm(formula, data = data),
    predictions = predictions,
    RMSE = sqrt(mean((data$sd_R - predictions)^2)),
    MAE = mean(abs(data$sd_R - predictions)),
    negative_predictions = sum(predictions < 0)
  )
}

log_evaluations <- lapply(log_formulas, evaluate_sd, data = train)
log_comparison <- imap_dfr(log_evaluations, function(result, name) {
  tibble(model = name, RMSE = result$RMSE, MAE = result$MAE,
         negative_predictions = result$negative_predictions)
}) |>
  arrange(RMSE)
knitr::kable(log_comparison, digits = 4)
```

model	RMSE	MAE	negative_predictions
log_both	5.9895	2.5211	0
log_SD	6.3570	2.6459	0
log_St	6.6788	5.7224	34
no_logs	6.7661	5.7872	34

```
best_log_name <- log_comparison$model[1]
best_additive <- log_evaluations[[best_log_name]]$fit
best_log_name
```

```
[1] "log_both"
```

Diagnostics for the best additive model

```
summary(best_additive)
```

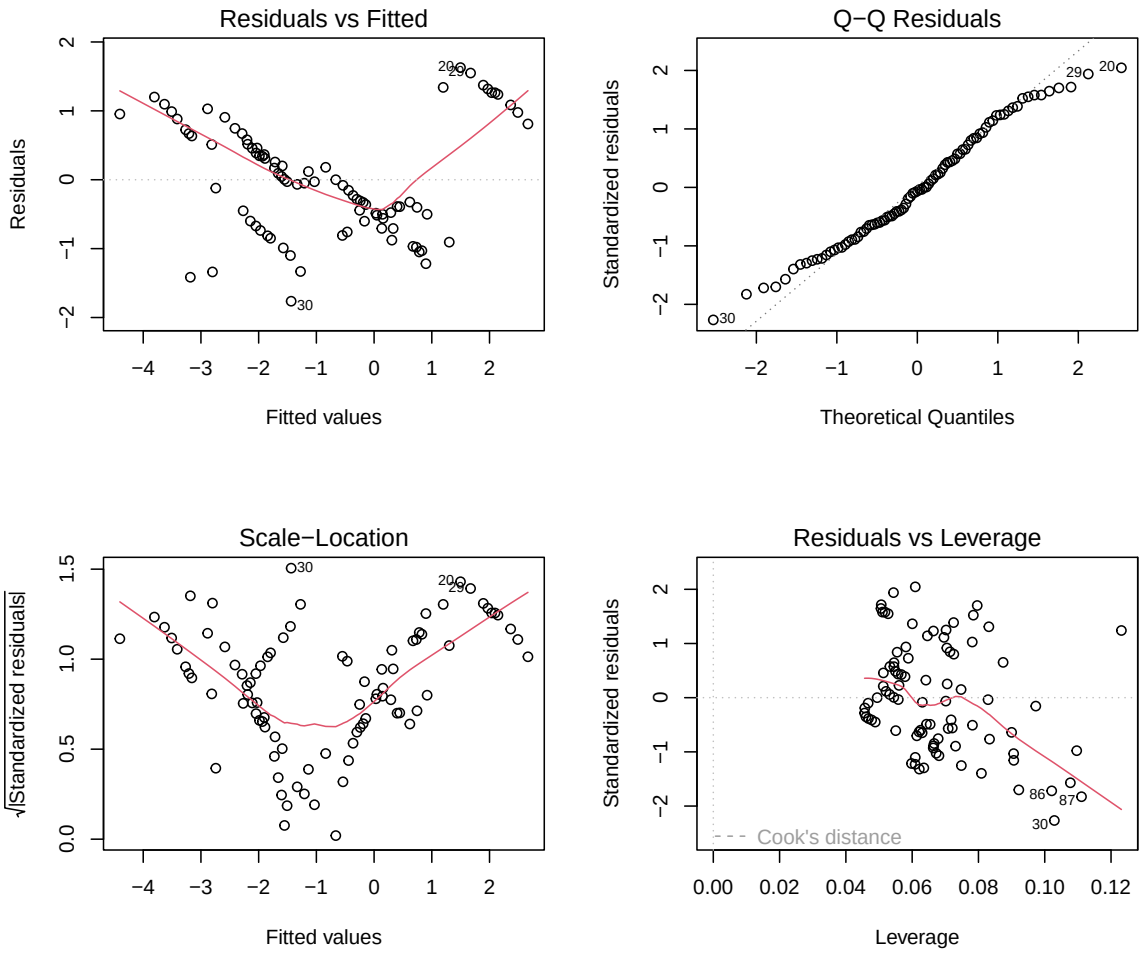
```
Call:
lm(formula = formula, data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-1.76349 -0.59978 -0.05008  0.63500  1.62809

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.1914     0.1906  11.496  < 2e-16 ***
log(St)       0.4317     0.0813   5.309 9.02e-07 ***
Re224        -2.3351     0.1977 -11.809  < 2e-16 ***
Re398        -3.9399     0.2488 -15.833  < 2e-16 ***
Fr0.3        -1.7468     0.2280  -7.663 3.04e-11 ***
FrInf        -1.3634     0.2035  -6.700 2.33e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8215 on 83 degrees of freedom
Multiple R-squared:  0.8152,    Adjusted R-squared:  0.804
F-statistic: 73.22 on 5 and 83 DF,  p-value: < 2.2e-16
```

```
par(mfrow = c(2, 2))
plot(best_additive)
```




```
par(mfrow = c(1, 1))
```

Compare interaction effects

```
# Build the interaction formulas using the selected log combination.
y_term <- if (best_log_name %in% c("log_SD", "log_both")) "log(sd_R)" else "sd_R"
x_term <- if (best_log_name %in% c("log_St", "log_both")) "log(St)" else "St"
```

```r
interaction_rhs <- c(
  additive = paste(x_term, "+ Re + Fr"),
  Re_Fr = paste(x_term, "+ Re * Fr"),
  St_Re = paste(x_term, "* Re + Fr"),
  St_Fr = paste(x_term, "* Fr + Re"),
  pairwise = paste0("(", x_term, " + Re + Fr)^2"),
  threeway = paste(x_term, "* Re * Fr")
)
interaction_formulas <- lapply(interaction_rhs, function(rhs) {
  as.formula(paste(y_term, "~", rhs))
})
interaction_evaluations <- lapply(interaction_formulas, evaluate_sd, data = train)
interaction_comparison <- imap_dfr(interaction_evaluations, function(result, name) {
  tibble(model = name, RMSE = result$RMSE, MAE = result$MAE,
         parameters = result$fit$rank,
         AIC = AIC(result$fit), BIC = BIC(result$fit),
         negative_predictions = result$negative_predictions)
}) |>
  arrange(RMSE)
knitr::kable(interaction_comparison, digits = 4)
```

model	RMSE	MAE	parameters	AIC	BIC	negative_predictions
Re_Fr	2.8986	0.9794	9	128.0299	152.9163	0
pairwise	3.9894	1.2496	13	126.1214	160.9623	0
St_Re	5.8076	2.4768	8	222.5563	244.9541	0
additive	5.9895	2.5211	6	225.3681	242.7885	0
St_Fr	6.1017	2.5868	8	229.2866	251.6844	0
threeway	6.9099	1.7744	16	126.3086	168.6154	0

```r
best_interaction_name <- interaction_comparison$model[1]
final_sd <- interaction_evaluations[[best_interaction_name]]$fit
final_sd_predictions <- interaction_evaluations[[best_interaction_name]]$predictions
formula(final_sd)
```

```
log(sd_R) ~ log(St) + Re * Fr
<environment: 0x1174c94a0>
```

Final-model summary and diagnostics

```
summary(final_sd)
```

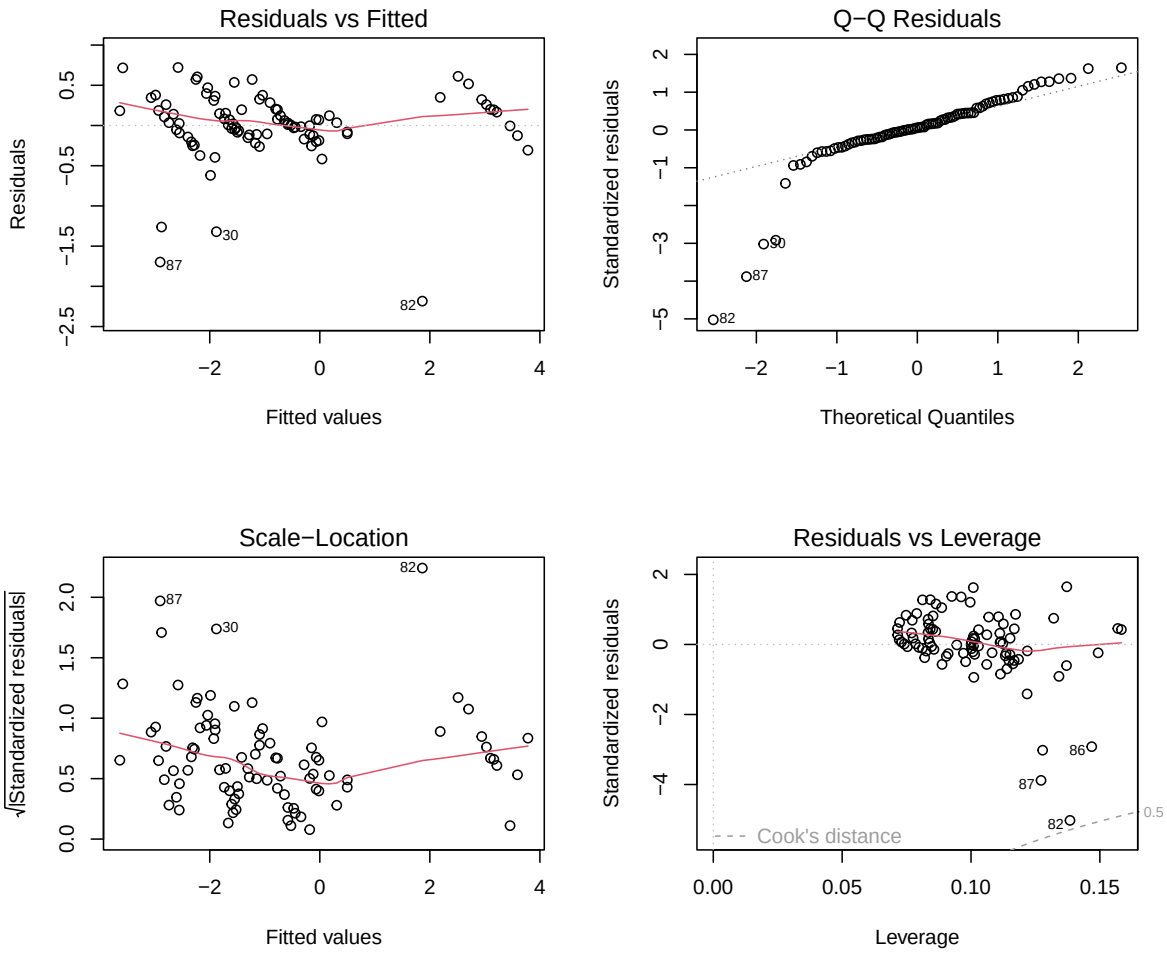
```
Call:
lm(formula = formula, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-2.1840 -0.1174  0.0254  0.2007  0.7211

Coefficients: (1 not defined because of singularities)
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.26776    0.13857  23.583  < 2e-16 ***
log(St)      0.46876    0.04685  10.005 9.43e-16 ***
Re224       -3.74284    0.18422 -20.317  < 2e-16 ***
Re398       -5.76886    0.20091 -28.714  < 2e-16 ***
Fr0.3       -3.28443    0.20090 -16.348  < 2e-16 ***
FrInf       -3.28295    0.20671 -15.882  < 2e-16 ***
Re224:Fr0.3  2.26260    0.27035   8.369 1.53e-12 ***
Re398:Fr0.3       NA         NA      NA       NA
Re224:FrInf  2.28606    0.27720   8.247 2.65e-12 ***
Re398:FrInf  3.55437    0.29842  11.911  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4683 on 80 degrees of freedom
Multiple R-squared:  0.9421,    Adjusted R-squared:  0.9363
F-statistic: 162.8 on 8 and 80 DF,  p-value: < 2.2e-16
```

```
par(mfrow = c(2, 2))
plot(final_sd)
```



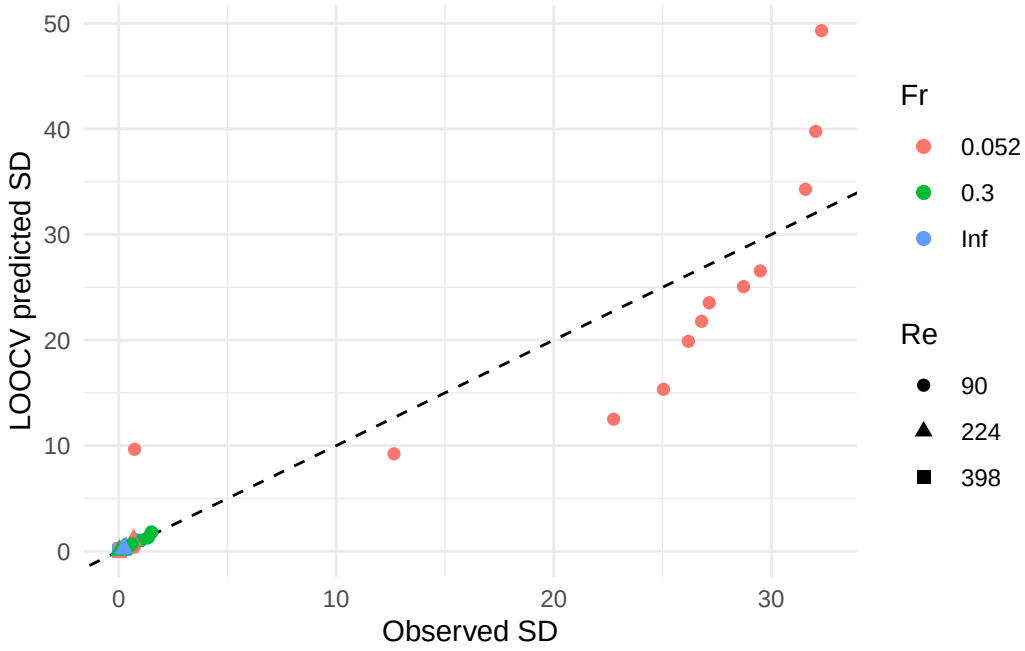

```
par(mfrow = c(1, 1))
```

Held-out predictions and largest errors

```
sd_predictions <- train |>
  mutate(row_id = row_number(), predicted_SD = final_sd_predictions,
         absolute_error = abs(sd_R - predicted_SD))

ggplot(sd_predictions, aes(sd_R, predicted_SD, color = Fr, shape = Re)) +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed") +
  geom_point(size = 2) +
```

```
  labs(x = "Observed SD", y = "LOOCV predicted SD") +
  theme_minimal()
```




```
sd_predictions |>
  slice_max(absolute_error, n = 5, with_ties = FALSE) |>
  select(row_id, St, Re, Fr, sd_R, predicted_SD, absolute_error) |>
  knitr::kable(digits = 4, caption = "Five largest LOOCV errors")
```

Table 3: Five largest LOOCV errors

row_id	St	Re	Fr	sd_R	predicted_SD	absolute_error
80	3.00	90	0.052	32.3153	49.3115	16.9962
20	0.20	90	0.052	22.7536	12.5080	10.2455
29	0.30	90	0.052	25.0451	15.3363	9.7088
82	0.05	90	0.052	0.7258	9.6549	8.9291
58	2.00	90	0.052	32.0449	39.7680	7.7231

Among the additive models, logging both SD and St gives the lowest LOOCV RMSE (about 5.99) and MAE (2.52). Keeping those transformations and adding a Re-Fr interaction reduces RMSE to about 2.90 and MAE to 0.98, the lowest values among the interaction candidates.

The selected model is `log(sd_R) ~ log(St) + Re * Fr`. It has no negative LOOCV predictions.

This suggests that the association between gravitational condition and SD differs across Reynolds-number levels. The estimated log-St coefficient is about 0.469, describing a positive adjusted association with SD; its slope is shared across conditions in this model. These are model-based associations.

The Q-Q plot shows lower-tail departures, and residual plots flag observations 82, 87, and 30 for investigation. The scale-location plot also suggests that residual spread is not fully constant. Keep all observations unless a data error or scientific reason justifies exclusion. Model selection improves predictive accuracy but does not remove these diagnostic limitations.